James R. Bounds, PhD

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EXECUTIVE SUMMARY

Experimental Physicist (PhD) and Hardware/Software Engineer with 10+ years of experience designing precision optical instruments, custom analog electronics, and scientific automation software. Expert in laser spectroscopy, frequency combs, precision metrology, and high-performance numerical computing.

TECHNICAL SKILLS

Optics & Photonics: Precision Free-Space Alignment, Optical Frequency Combs, Cavity-Enhanced Spectroscopy (CRDS, Vernier), Dual-Comb Spectroscopy, Optical Fiber Splicing & Coupling, Interferometry, Wavelength Metrology, Multi-Pass Cells.

Hardware, EDA & Electronics: Analog/Digital/Mixed-Circuit Design, Schematic Capture & PCB Layout (KiCAD), Circuit Simulation (NI Multisim, QUCS), FPGA Fabric Design (Verilog), Microcontrollers, Photodetector Optimization, PID Controllers, VNAs, Oscilloscopes.

Instrumentation & DAQ: Hardware Protocol Automation (GPIB, RS232, USB, Ethernet, SPI, I^2C, PyVISA), Custom DAQ Systems, FPGA Acceleration, GUI Development (Qt/PyQt, Tkinter, LabVIEW).

Programming Languages: Python, C, C++, Cython, Verilog, MATLAB, Java, Bash, Shell Scripting, Git, LaTeX.

Scientific Computing & Modeling: NumPy, SciPy, PyTorch, Numba, OpenCV, Scikit-learn, Pandas, Slurm HPC, Ray-Tracing & Gaussian Beam Modeling.

Mechanical & CAD/CAM: 3D CAD Modeling (Autodesk Inventor, Blender, PyVista), Precision Machining (Vertical Mill, Precision Lathe), 3D Printing, Ultra-High Vacuum (UHV) Systems.

Systems & Infrastructure: Linux Administration (openSUSE, Debian, Arch, Fedora, RHEL), SSH, Networking, X11 Forwarding.

RESEARCH & ENGINEERING EXPERIENCE

Graduate Research Assistant / Teaching Assistant **2016 – 2026**
Texas A&M University *College Station, TX*

- **Precision Optical Setup Design:** Engineered, assembled, and aligned complex free-space optical systems—including Cavity Ring-Down Spectroscopy (CRDS), Vernier spectroscopy, and Doppler-free saturated absorption setups—achieving sub-ppb sensitivity.
- **Laser Characterization & Frequency Metrology:** Calibrated and characterized mid-IR and near-IR laser sources using optical frequency combs, custom wavemeters, and interferometers to measure hyperfine spectra with sub-MHz precision.
- **Optical & Beam Simulation:** Developed custom Gaussian beam and ray-tracing simulation GUIs in Python and C to model beam propagation, mode matching, and cavity coupling efficiency.
- **Optoelectronics & PCB Engineering:** Executed full life-cycle development of low-noise, high-bandwidth photodetectors and signal-conditioning electronics from schematic capture to KiCAD PCB layout, fabrication, and testing.
- **Instrument Automation & DAQ Stack:** Architected full-stack automation pipelines in Python (PyVISA, PyQt) and C, integrating FPGA fabric design for real-time data acquisition and high-speed processing.
- **Optomechanics & Fabrication:** Designed custom mounts and vacuum interface enclosures using CAD (Autodesk Inventor), 3D printing, and hands-on precision machining (lathe/milling).
- **Fiber Optics & Assembly:** Handled optical fiber preparation, fusion splicing (single-mode and PM fiber), and free-space-to-fiber coupling with loss minimization across visible and mid-IR wavelengths.

- **Error Quantification & Optimization:** Developed non-linear fitting algorithms and numerical optimization routines to validate raw spectral data against molecular databases (e.g., HITRAN).

Visiting Scientist (Summer Terms) **2016 – 2025**
Max Planck Institute of Quantum Optics *Garching, Germany*

- Collaborated with international teams under Prof. Thomas Udem on frequency comb metrology and spectroscopy.
- Developed and tested hardware prototypes and presented findings on tight project timelines.

IT Technician **2015 – 2016**
Systek Computing *College Station, TX*

- Configured Linux and Windows server environments, network infrastructure, and system diagnostics.
- Both in-store and on-site customer service.

PROJECTS & HARDWARE SYSTEMS

- **Optical Breath Analysis Platforms:** Designed and aligned CRDS and Vernier spectrometers capable of sub-1ppb trace gas detection in human breath samples.
- **Wavelength Metrology Platform:** Built a real-time wavelength analysis instrument achieving precision $< 10^{-8}$ using novel data processing algorithms.
- **Precision IR Spectrometer:** Designed and implemented a saturated absorption spectrometer with frequency comb calibration to measure metastable xenon hyperfine structures.
- **Open Source Hardware/Software Projects:**
  - **Battery-Powered Amplified Photodetector:** Low-noise PCB optoelectronic design (GitHub).
  - **Gaussian Beam Simulator GUI:** Qt/Python optics simulation tool (GitHub).

EDUCATION

PhD in Physics **2026**
Texas A&M University *Advisor: Prof. Alexei Sokolov*
B.S. in Physics **2015**
Texas A&M University *Department of Physics and Astronomy*

SELECTED PUBLICATIONS & PRESENTATIONS

**Full list of 9 peer-reviewed journal papers, 6 conference proceedings, and 5 talks available upon request.*

- **Bounds, J.**, et al. "Acetone Absorption Cross-Section in the Near-Infrared of the Methyl Stretch Overtone and Application for Analysis of Human Breath." *Optics* 6 (1), 2025.
- **Bounds, J.**, et al. "Hyperfine structure and isotope shifts of xenon measured for near-infrared transitions with Doppler-free saturated absorption spectroscopy" *Spectrochimica Acta Part B: Atomic Spectroscopy*, 202, 2023.
- Ozawa, A., Davila-Rodriguez, J., **Bounds, J. R.**, et al. "Single ion fluorescence excited with a single mode of an UV frequency comb." *Nature Communications* 8 (1), 2017.
- Zhu, F., **Bounds, J.**, et al. "Near infrared frequency comb vernier spectrometer for broadband trace gas detection." *Optics Express* 22 (19), 2014.
- **Conference Talk (2024):** "Frequency Comb Calibration of the Hyperfine Structure of Metastable Xenon," *SIBOR Workshop*, Montgomery, TX.